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RESEARCH ARTICLE



DML-Geo: an ensemble double machine learning framework for estimating spatially heterogeneous causal effects

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ABSTRACT

Causal inference in geographical sciences faces the challenge of isolating treatment effects from high-dimensional observational data, complicated by spatial non-stationarity and persistent confounding. Double machine learning (DML) offers a powerful solution for high-dimensional debiasing through orthogonalization and cross-fitting, but traditional variants overlook spatial heterogeneity by treating space as a simple covariate. To address this, we introduce DML-Geo, an ensemble extension of DML for estimating spatially varying causal effects. Retaining the orthogonalization procedure of DML at its first stage, DML-Geo augments the second stage with three complementary estimators, namely a linear regression model for covariate-driven effects, a generalized additive model (GAM) for spatially smoothed additive effects, and geographically weighted regression (GWR) for localized patterns. Robustness is further enhanced by an adaptive weighting scheme based on inter-model correlations to aggregate outputs from these variants, complemented by a bootstrap procedure for significance testing. Extensive simulations confirm DML-Geo's superior precision and stability relative to its component models and competing baselines. In real-world applications to housing prices and mental health outcomes, DML-Geo uncovers interpretable spatial causal effect patterns, offering place-specific insights to support policy decisions. DML-Geo provides a flexible toolkit for geospatial causal inference that does not require causal graphs or strong structural assumptions.

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1. Introduction

In geographical sciences, causal research has increasingly moved beyond correlative patterns to uncover the mechanisms driving social and environmental processes (Chen *et al.* 2021, Fu *et al.* 2025), such as the effects of urban green space on public health or the impact of land-use change on air quality. This shift reflects a broader recognition that causal analysis is essential for attributing observed outcomes to specific drivers and for anticipating how systems respond to interventions or policy change. Yet establishing causal relationships in geographical sciences is inherently difficult (Zhang and Wang 2025, Zhang *et al.* 2026). Observational data are spatially extensive and shaped by intertwined geographic factors, and the counterfactual nature of causal inference makes it impossible to simultaneously observe a spatial unit under both treated and control conditions.

Double Machine Learning (DML) offers a principled solution to the challenges of estimating causal effects from high-dimensional and confounded observational data (Chernozhukov *et al.* 2018). By integrating the flexibility of machine learning with econometric debiasing techniques, DML yields asymptotically normal estimators under the assumption of unconfoundedness. Central to this framework is the Neyman orthogonality condition, which makes the causal effect estimate insensitive to first-order errors in the nuisance models, thereby preventing such errors from biasing the final inference. DML further employs cross-fitting, a sample-splitting procedure in which nuisance models are trained on one data subset and evaluated on the remaining subsets, to mitigate overfitting. These properties make DML particularly well-suited to high-dimensional geospatial data, where a large number of covariates (e.g., remote sensing metrics or topographic attributes) would otherwise complicate causal inference.

While DML has gained widespread adoption across domains (Fuhr *et al.* 2024, Ling *et al.* 2024, Jiang and Sun 2025), spatial heterogeneity remains largely overlooked in classical DML frameworks. Spatial heterogeneity denotes the location-dependent variation in relationships between variables, where processes differ systematically across space due to contextual factors. As a foundational principle in geography (Goodchild 2004), this implies that causal relationships among geographic phenomena are likely to exhibit spatial non-stationarity. For example, the impact of identical urban environmental factors on the urban heat island effect may vary across different local climate zones (LCZs) at distinct spatial locations (Zhong *et al.* 2024). However, although previous studies acknowledge this issue, they often treat spatial elements, such as coordinates or spatial lags, as ordinary covariates within machine learning pipelines, without explicitly modeling the underlying spatial structure (Credit and Lehnert 2024, Gilbert *et al.* 2024). Hence, improving the adaptivity of causal inference models, under the assumption that spatial causal heterogeneity may exist, emerges as a critical prerequisite for yielding reliable estimates in geospatial contexts.

To address these problems, we present DML-Geo, a novel ensemble framework that extends DML to geospatial settings by integrating three complementary second-stage estimators: linear regression, generalized additive models (GAMs), and geographically weighted regression (GWR). These yield three variants: DML-CATE, which captures covariate-driven heterogeneity via linear regression; DML-GAM, which combines additive covariate effects with moderate spatial smoothing; and DML-GWR, which employs

GWR for localized spatial causal estimation. To ensure robustness regardless of whether spatial heterogeneity is present, the three variants are aggregated in DML-Geo through a correlation-based weighting scheme, with bootstrap-based significance testing incorporated for inference. DML-Geo directly tackles two key questions: (1) How can spatial localization be incorporated into DML to model geographically varying causal effects? (2) How can an ensemble design ensure precision and generalizability when spatial causal heterogeneity may or may not dominate? The proposed framework provides a flexible and efficient toolkit for causal inference with high-dimensional geographic data without requiring elaborate causal graphs or domain-specific structural assumptions.

The rest of the article unfolds as follows: [Section 2](#) details the methodological integration and derivations; [Section 3](#) benchmarks our method against alternative models and evaluates its robustness via extensive simulations; [Section 4](#) demonstrates empirical applications to real geospatial datasets; [Section 5](#) discusses implications and limitations; and [Section 6](#) concludes this study.

2. Related works

Causal inference lies at the heart of empirical research, enabling researchers to move beyond associations and identify what actually drives observed outcomes (Pearl 2010). Causal inference is fundamentally different from correlation analysis or post-hoc interpretability methods such as SHAP (SHapley Additive exPlanations) (Chen *et al.* 2023, Naser 2024). Correlation captures co-occurrence but fails to imply directionality or intervention effects. For example, rain and umbrella sales rise together, yet umbrellas do not cause rain. Causal methods, by contrast, invoke counterfactual reasoning to ask ‘what if’ questions under changed conditions, guarding against spurious associations via confounding control. SHAP provides post-hoc explanations of predictions from machine learning models. While Shapley values can admit a causal interpretation under restrictive assumptions, they generally reflect associative rather than causal pathways, explaining how a model allocates predictions rather than why an effect occurs (Lundberg and Lee 2017, Chen *et al.* 2023). This distinction is important in applied settings, where conflating attribution with causation can mislead policy or prediction.

Broadly, two major paradigms shape the field of causal inference. The first, rooted in structural causal models (SCMs) and directed acyclic graphs (DAGs), draws from Pearl’s graphical approach, where interventions are simulated by severing edges in a causal diagram to reveal mechanisms and counterfactuals (Pearl 2010). This school emphasizes identifiability through backdoor criteria or front-door adjustments, powering methods like instrumental variables and mediation analysis (Pearl and Mackenzie 2018). The second paradigm, namely the potential outcomes framework (POF, also known as the Rubin causal model) (Rubin 1978), treats causality as a comparison of what would happen under different treatments for the same unit, with estimation relying on assumptions like conditional exchangeability or unconfoundedness. Representative techniques under POF include propensity score matching (Rosenbaum and Rubin 1983), Meta-Learners (Künzel *et al.* 2019), tree-based approaches like causal forest (Wager and Athey 2018), and deep-learning-based models like CFRNet (Shalit *et al.* 2017). These methods

have significantly advanced causal inference across diverse scenarios, though challenges remain in handling high-dimensional data and diagnosing spatial causal relationships (Yao *et al.* 2021, Gao *et al.* 2023, Credit and Lehnert 2024).

Within POF, DML distinguishes itself through an integration of high-dimensional nuisance parameter estimation and orthogonal scoring (Chernozhukov *et al.* 2018). Leveraging cross-fitting and Neyman orthogonality, it ensures reliable recovery of treatment effects even when confounders are flexibly modeled using machine learning algorithms such as random forests. A key practical strength is its ability to accommodate high-dimensional covariate arrays without succumbing to the curse of dimensionality, while requiring only an initial delineation of treatments (T), outcomes (Y), and confounders (X). This is particularly advantageous for exploratory research in geographical sciences, where datasets often comprise diverse variables like remote sensing indices, climatic proxies, or topographic metrics (Yan and Roy 2015, Berkessa *et al.* 2025). DML has also been extended to address spatial confounding. Gilbert *et al.* (2024) showed that, under the assumption that unmeasured confounders are measurable functions of space, treating spatial coordinates as proxies allows DML to eliminate confounding bias without restrictive parametric assumptions, enabling valid causal effect estimation in complex spatial settings.

A pivotal advancement in causal analysis is the estimation of the Conditional Average Treatment Effect (CATE), which captures how causal impacts differ across subgroups or contexts. In geographical settings, treatment effects are likely to vary across both covariate subgroups and spatial locations. Yet, existing causal machine learning frameworks tend to handle spatial information in a relatively straightforward manner, incorporating it as ordinary input features. For instance, Credit and Lehnert (2024) incorporate spatial lag terms into causal forests and T-learners to account for spatial autocorrelation and spillover effects, and their approach has demonstrated utility in settings where such spillovers are prominent. However, incorporating spatial information as ordinary input features leaves spatial heterogeneity in treatment effects to be learned implicitly. As a result, spatial estimates may vary across learners, depend on tuning, and risk capturing noise instead of genuine geographic patterns.

GWR is a well-established approach in geographical sciences for explicitly modeling spatial heterogeneity through localized kernel weighting, allowing relationships between variables to vary across space (Brunsdon *et al.* 1996). Theoretically, GWR's localized estimation aligns naturally with DML: embedding GWR in the second stage would allow DML to retain its confounding adjustment while recovering geographically varying treatment effects through spatial weighting. However, the effectiveness of this hybrid approach remains largely unexplored. Furthermore, causal heterogeneity is inherently unobservable due to the counterfactual nature of potential outcomes (Qi *et al.* 2021). This uncertainty underscores the need for robust model designs that perform reliably across scenarios, avoiding overfitting to spurious spatial patterns when true heterogeneity is minimal or absent.

3. Methodology

This study introduces DML-Geo, a causal inference framework designed to estimate spatially varying treatment effects in the presence of high-dimensional confounding.

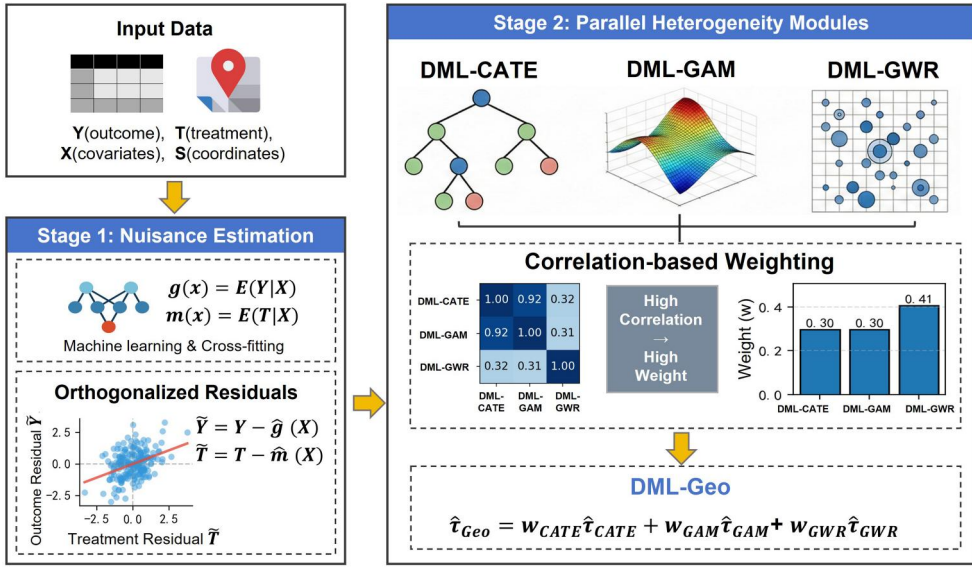


Figure 1. Conceptual diagram of DML-Geo.

As shown in Figure 1, the core idea is to retain DML's orthogonalization to remove confounding bias, while adding a parallel heterogeneity module in the second stage. This module encompasses three complementary models (linear regression model, GAM, and GWR) to capture non-spatial, spatially smooth, and spatially nonstationary forms of treatment effect variation. Their outputs are then combined through an adaptive weighting scheme based on their correlation coefficients, producing a unified causal effect surface that integrates both attribute-driven variations and spatial structures.

3.1. Foundations of DML

DML operates in the potential outcomes setting, with the assumption that the data arise from a structural mechanism as:

$$\begin{cases} Y = \tau T + g(X) + \varepsilon \\ T = f(X) + \eta \end{cases} \text{ with } E[\varepsilon|X] = 0, E[\eta|X] = 0 \quad (1)$$

where Y represents the outcome, X the covariates, and T the treatment (or variable of interest). τ represents the causal effect, i.e., the ATE.

DML fundamentally draws on the Frisch–Waugh–Lovell (FWL) theorem to estimate τ (Frisch and Waugh 1933). The theorem shows that, in a linear regression $Y = \beta T + \gamma X + \epsilon$, β is equivalent to the coefficient obtained by residualized outcome (i.e., $Y - E[Y|X]$) on the residualized treatment (i.e., $T - E[T|X]$). This procedure effectively isolates the partial effect of T on Y by eliminating confounding from X .

Extending FWL to nonlinear settings, DML employs flexible machine learning models (e.g., random forests) to estimate the nuisance functions $g(X) = E[Y|X]$ and $f(X) = E[T|X]$ in Equation (1), while preserving the orthogonalization that ensures reliable causal estimates. Notably, the nuisance functions must be estimated using

cross-fitting, which is essential to avoid overfitting and to guarantee that the estimator of τ remains unbiased and valid for inference (Chernozhukov *et al.* 2018). Following the nuisance estimation, the original DML performs a simple linear regression of the orthogonalized residuals, which are defined as the outcome residual $\tilde{Y} = Y - \hat{g}(X)$ and the treatment residual $\tilde{T} = T - \hat{m}(X)$, to obtain the ATE estimate $\hat{\tau}$.

To accommodate heterogeneity, DML extends to CATE by replacing the constant τ with a covariate-dependent function $\tau(X)$, which can be estimated by solving the following linear equation:

$$\tilde{Y} = \tau(X)\tilde{T} + \varepsilon \quad (2)$$

To distinguish in subsequent sections whether causal effects account for heterogeneity, the global ATE version is termed DML, while the heterogeneous extension is DML-CATE.

3.2. Incorporating GWR for spatial heterogeneity in DML

GWR is widely employed for modeling spatial non-stationarity by estimating location-specific parameters through weighted least squares at each observation point. The model fits a local regression $Y = \beta(u, v)X + \epsilon$, where coefficients $\beta(u_i, v_i)$ at point (u_i, v_i) are derived as $\beta(u_i, v_i) = (X^T W_i X)^{-1} X^T W_i Y$. Let d_{ij} denote the distance between locations i and j . The weight W_{ij} is defined by a kernel function with a bandwidth parameter b_i , which controls the spatial scale of the local calibration. In this study, we employ a bisquare kernel:

$$W_{ij} = \begin{cases} \left(1 - \left(\frac{d_{ij}}{b_i}\right)^2\right)^2, & d_{ij} < b_i \\ 0, & d_{ij} \geq b_i \end{cases} \quad (3)$$

and adopt an adaptive bandwidth scheme in which b_i varies spatially to accommodate uneven sampling density. The optimal bandwidth is determined by minimizing the corrected Akaike Information Criterion (AICc), solved efficiently via a golden-section search, a standard approach in GWR bandwidth calibration.

GWR is incorporated into the second stage of DML, replacing the global regression with localized estimation while keeping the first-stage nuisance model unchanged. This design is intentional: incorporating GWR earlier could distort the nuisance functions and compromise the orthogonality condition, whereas applying it after orthogonalization allows the framework to capture spatially varying effects without interfering with the debiasing step (Chernozhukov *et al.* 2018, Oprescu *et al.* 2019). Consequently, embedding these orthogonalized residuals into GWR yields a spatially varying CATE, where the causal effect at location i is estimated as $\hat{\tau}(u_i, v_i) = (\tilde{T}^T W_i \tilde{T})^{-1} \tilde{T}^T W_i \tilde{Y}$.

3.3. Bridging spatial and attribute heterogeneity: DML-GAM and ensemble integration

While DML-GWR effectively addresses spatial non-stationarity by deriving location-dependent coefficients, it focuses exclusively on positional heterogeneity, assuming

uniform effects from covariates across space. DML-CATE, by contrast, captures covariate-driven heterogeneity but does not account for geographic variation. In this context, GAMs offer a natural intermediate solution. As noted by Comber *et al.* (2024), GWR can be regarded as a particular instance of a GAM, and both reduce to linear regression when spatial and nonlinear components are removed, reflecting the same specification underlying DML-CATE. However, unlike GWR, which estimates independent local models through a moving-window approach, GAM captures both spatial and covariate-driven variation through smooth functions within a unified global framework (Fan and Huang 2022, Comber *et al.* 2024, Zhang *et al.* 2025). As GAM simultaneously captures covariate-driven and spatial heterogeneity, it serves as a natural complement to linear regression and GWR.

On this basis, this study formulates DML-GAM by embedding GAM into the second stage of the DML framework. DML-GAM imposes a tensor product smooth $te(u, v)$ over spatial coordinates (u, v) to accommodate location-dependent nonlinearities, and univariate smooth functions $s_j(X_j)$ to capture the additive effects of other covariates. The orthogonalized residuals $\tilde{Y} = Y - \hat{g}(X)$ and $\tilde{T} = T - \hat{m}(X)$ are then linked through:

$$\tilde{Y} = \tau(X, u, v)\tilde{T} + \varepsilon \quad (4)$$

where the heterogeneous treatment effects are approximated by a GAM decomposition:

$$\tau(X_i, u_i, v_i) \approx \sum_{j=1}^p s_j(X_{ij}) + te(u_i, v_i) \quad (5)$$

This hybrid CATE estimate $\hat{\tau}(X_i, u_i, v_i)$ from DML-GAM blends spatial and covariate dynamics, positioning itself as an intermediate diagnostic model between the location-centric DML-GWR and the attribute-oriented DML-CATE.

The three variants capture complementary forms of effect heterogeneity, but relying on any single model risks misspecification because the appropriate form of spatial heterogeneity is unknown in advance. Ensemble integration is therefore advisable to harness their complementary strengths while mitigating individual model bias. A common solution is to weight models by predictive performance, as adopted in uncertainty-based ensemble approaches such as Bayesian Model Averaging (Hoeting *et al.* 1999), but this requires held-out outcome observations that are fundamentally unavailable in causal inference, where the treatment effect $\tau(x)$ is inherently unobservable for any unit. To that end, this study adopts a correlation-based weighting scheme in which each model's weight is proportional to its mean pairwise correlation with the remaining two models, such that models with higher inter-model agreement receive greater weight. This approach requires no ground-truth observations and is conceptually analogous to majority voting in classification, extended to the continuous domain. Similar weighting strategies have also been applied in prior ensemble learning studies (Lai *et al.* 2021, Deng *et al.* 2021).

Let $\hat{\tau}_1$, $\hat{\tau}_2$, and $\hat{\tau}_3$ denote the CATE vectors from the models DML-CATE, DML-GAM, and DML-GWR. The Pearson correlation matrix C is first computed across these vectors, with off-diagonal elements $\rho_{kl} = \frac{\text{Cov}(\hat{\tau}_k, \hat{\tau}_l)}{\sigma_k \sigma_l}$ for model k and l . Notably, any negative ρ_{kl} values are truncated to zero. For each model k , the average pairwise correlation is:

$$\bar{\rho}_k = \frac{1}{M-1} \sum_{l \neq k} \rho_{kl} \text{ (with } M = 3\text{)}. \quad (6)$$

Weights are normalized as $w_k = \frac{\bar{\rho}_k^\alpha}{\sum_k \bar{\rho}_k^\alpha}$ to ensure $\sum_k w_k = 1$, where α is a parameter to modulate sensitivity to correlation disparities; when all $\bar{\rho}_k = 0$, uniform weights (i.e., $w_k = 1/3$) are assigned. The ensemble CATE is then expressed as:

$$\hat{\tau}_{\text{ens}}(X_i, u_i, v_i) = \sum_k w_k \hat{\tau}_k(X_i, u_i, v_i) \quad (7)$$

This scheme adaptively reflects empirical patterns: high agreement between DML-CATE and DML-GAM ($\rho_{\text{CATE}, \text{GAM}}$) elevates their weight w_{CATE} and w_{GAM} , highlighting the importance of attribute effects while reducing the role of spatially isolated patterns in DML-GWR; high agreement between DML-GWR and DML-GAM ($\rho_{\text{GWR}, \text{GAM}}$) shifts weight toward spatial components w_{GAM} and w_{GWR} , thereby prioritizing locational variance. The resulting framework, referred to as DML-Geo throughout this paper, balances the strengths of all three variants and remains robust across diverse geographical settings where multiple forms of heterogeneity may coexist.

3.4. Evaluating the significance of ensemble estimates via placebo tests

Significance for individual models (DML-CATE, DML-GAM, DML-GWR) is evaluated by randomizing the treatment variable via bootstrap. This allows evaluation of whether the observed estimates deviate meaningfully from chance (van der Laan *et al.* 2025).

For each placebo iteration $k = 1, \dots, K$, a permuted treatment vector $D^{(k)}$ is generated by randomly shuffling D . Then, the three models are refitted using $D^{(k)}$ to produce $\hat{\tau}_{\text{cate}}^{(k)}, \hat{\tau}_{\text{gam}}^{(k)}, \hat{\tau}_{\text{gwr}}^{(k)}$. P-values are computed as $p_i = \frac{1}{K} \sum_{k=1}^K \mathbb{I}(|\hat{\tau}_i^{(k)}| \geq |\hat{\tau}_i|)$. For DML-Geo, the observed $\hat{\tau}_{\text{geo}} = \sum_{i=1}^3 w_i \hat{\tau}_i$ is contrasted against placebo ensembles $\hat{\tau}_{\text{geo}}^{(k)} = \sum_{i=1}^3 w_i^{(k)} \hat{\tau}_i^{(k)}$. The p -value follows as $p = \frac{1}{K} \sum_{k=1}^K \mathbb{I}(|\hat{\tau}_{\text{geo}}^{(k)}| \geq |\hat{\tau}_{\text{geo}}|)$.

4. Experiments and analysis

4.1. Benchmarking on simulated data

To empirically validate the proposed DML-Geo framework and its components, we benchmark them against GeoSHAPLEY (Li 2024a), Spatial T-learner (STL) (Credit and Lehnert 2024) and CausalForest (Athey and Wager 2019). GeoSHAPLEY is a game-theoretic method that extends the Shapley value framework to quantify feature importance in machine learning models while explicitly accounting for spatial dependence. GeoSHAPLEY evaluates both covariate contributions and location-aware attributions to model predictions, making it conceptually comparable to our approach. It should be noted, however, that GeoSHAPLEY targets associational explanations, whereas the DML family is designed to recover causal effects. STL adapts the T-learner framework to geographic settings by training separate outcome models for treated and control groups, with spatially lagged variables included as inputs to account for spatial dependence. CausalForest, in turn, adapts random forests to estimate heterogeneous treatment effects by recursively partitioning the covariate space, allowing effects to

vary flexibly across subgroups without parametric assumptions (Athey and Wager 2019). This tree-based approach shares DML's focus on conditional average treatment effects but relies on splitting criteria to capture nonlinearity, providing a non-spatial baseline for comparison with DML-Geo.

4.1.1. Experimental setup

This experiment aims to compare GeoSHAPLEY, STL and CausalForest against DML in terms of their ability to identify the causal effects of covariates X on outcome Y , that is, the coefficients of X . Since the intercept term carries no causal interpretation, it was excluded from the comparison despite being estimable by GeoSHAPLEY. In addition, as DML variants estimate treatment-specific τ , multiple runs were performed for CausalForest, STL and the proposed DML-Geo by designating each predictor sequentially as the treatment variable (with remaining X as confounders), yielding the causal effects for all covariates on Y . GeoSHAPLEY, by contrast, provides attributions for all features in a single fit.

Three simulated datasets were designed in the following experiments. The spatial domain was defined as a 50×50 grid, where the coordinates (u, v) ranged from 0 to 49, yielding a total of 2,500 georeferenced observations. Each dataset was generated using predefined mathematical equations, with the coefficients assigned to each variable serving as the ground truth treatment effects. This design allows the recovery of causal signals to be rigorously evaluated by comparing estimated effects against known true values. Notably, the parameter α in DML-Geo was set to 10 in this simulation experiment, while its sensitivity will be further examined in Section 4.2.3.

Dataset 1 corresponds to a synthetic dataset introduced in the original GeoSHAPLEY paper, which introduces spatially varying coefficients generated from Gaussian Random Fields (GRFs). Two spatial processes with distinct smoothness parameters (length scales of 5 and 15) were used to produce coefficient surfaces $b_1(u, v)$ and $b_2(u, v)$. The response variable was defined as:

$$Y = b_0 + b_1(u, v)X_1 + b_2(u, v)X_2 + \varepsilon \quad (8)$$

where $b_0 = 3$ and $\varepsilon \sim N(0, 1)$. This design generates spatially heterogeneous effects and serves to evaluate the models' ability to recover spatially nonstationary relationships.

Dataset 2 corresponds to another synthetic dataset introduced in the original GeoSHAPLEY study. The outcome variable Y was generated from a nonlinear model involving four predictors X_1, X_2, X_3, X_4 , which was defined as:

$$Y = b_0(u, v) + b_1(u, v)X_1 + b_2(u, v)X_2 + b_3X_3 + b_4X_4^2 \quad (9)$$

where $b_0(u, v) = \frac{6}{12^4} \left[12.5^2 - \left(12.5 - \frac{u}{2} \right)^2 \right] \left[12.5^2 - \left(12.5 - \frac{v}{2} \right)^2 \right]$, $b_1(u, v) = \frac{3(u+v)}{49}$, $b_2(u, v) = \frac{3(49-u+v)}{49}$, $b_3 = 2$, and $b_4 = 1$. This configuration defines a smooth spatially varying intercept $b_0(u, v)$ and spatial gradients in $b_1(u, v)$ and $b_2(u, v)$, while X_3 and X_4^2 contribute globally constant effects. By introducing the quadratic term $b_4X_4^2$, we simulated the process where the marginal effect of X_4 varies with its own values (i.e., attribute-space non-stationarity).

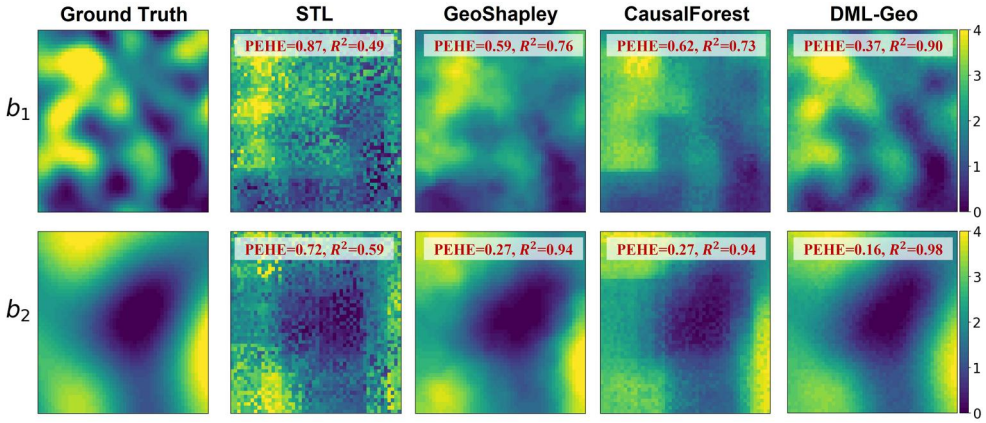


Figure 2. Comparison of estimated coefficients for Dataset 1 from different models. Results from GeoShapley are post-processed by GWR smoothing.

Dataset 3 was modified from Dataset 2 with additional structural modifications. Specifically, the original component b_3X_3 was replaced by interaction effects $b_3X_1X_3$, while the intercept term $b_0(u, v)$ was incorporated as the coefficient of a new covariate X_0 . As a result, the outcome variable Y was defined as:

$$Y = b_0(u, v)X_0 + b_1(u, v)X_1 + b_2(u, v)X_2 + b_3X_1X_3 + b_4(u, v)X_4^2 \quad (10)$$

Building on the basic structure of Dataset 2, this design introduces an interaction term $b_3X_1X_3$ and more complex spatial effects, including a spatially varying term $b_0(u, v)X_0$ and a nonlinear spatially varying term $b_4(u, v)X_4^2$, thereby constructing a more challenging scenario for spatial causal estimation.

Model performance is assessed using two metrics: the coefficient of determination R^2 , which quantifies explained variance between true and estimated effects, and Precision in the Estimation of Heterogeneous Effects (PEHE), which is defined as $\sqrt{E[(\hat{\tau}(X) - \tau(X))^2]}$ to capture the overall error between estimated and true heterogeneous effects.

4.1.2. Results

For Dataset 1, GeoSHAPLEY was implemented using the TabNet model following the original study (Li 2024a) to estimate the coefficients b_1 and b_2 . As illustrated in Figure 2, STL exhibits limited performance, with pronounced pixelated noise in the estimated surfaces and the lowest PEHE and R^2 values across all datasets. Although GeoSHAPLEY and CausalForest capture broadly similar trends, they display evident spatial discrepancies compared to the ground truth, including over-smoothed hotspots and distorted local effects. In contrast, DML-Geo substantially outperforms others, attaining lower PEHE values and delivering consistently higher R^2 values over 0.9, especially for coefficient b_1 , which exhibits a more complex spatial pattern than b_2 in this example. This finding highlights the superior capability of the DML-Geo framework in detecting complex spatial heterogeneity in causal effects.

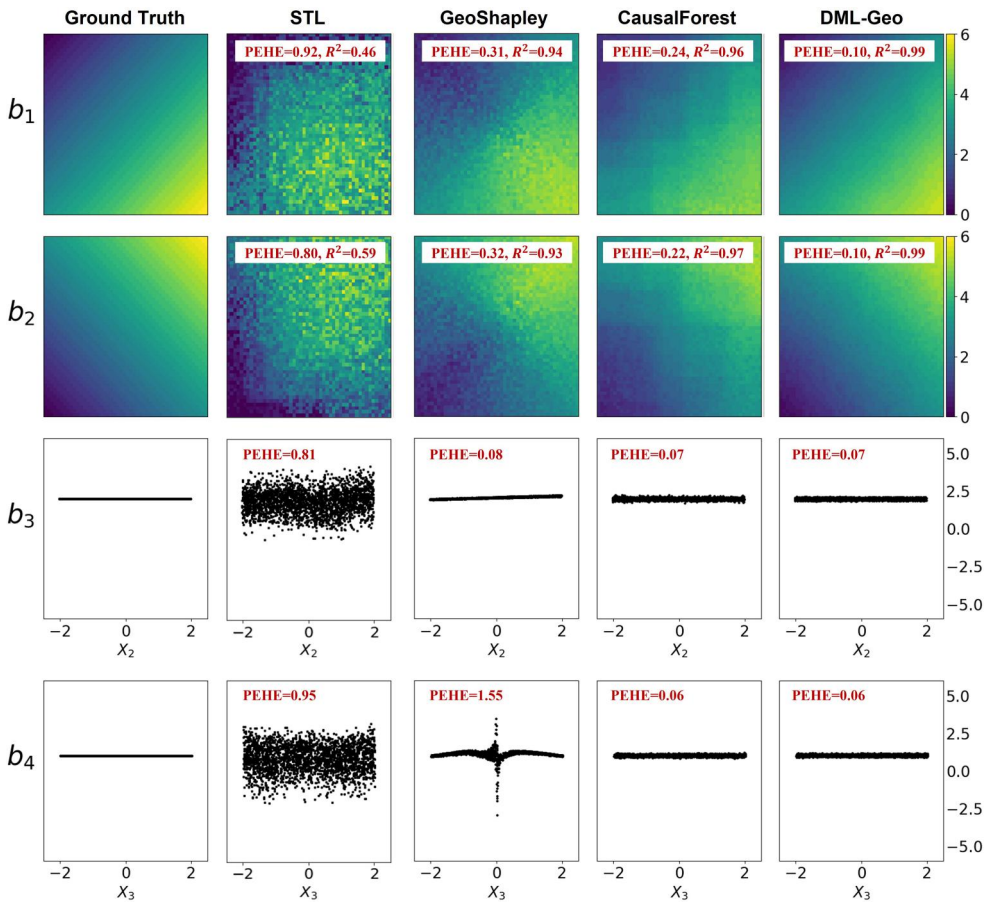


Figure 3. Comparison of estimated coefficients for Dataset 2 from different models. Notably, R^2 is undefined when the ground truth is constant.

For Dataset 2, GeoSHAPLEY was implemented using the neural network (NN) model, identified as one of the best-performing configurations in the original study (Li 2024a). As illustrated in Figure 3, STL fails to produce stable estimates, exhibiting severe noise across all coefficients and the lowest R^2 values. In contrast, DML-Geo, CausalForest, and GeoSHAPLEY successfully recover the coefficients b_1 and b_2 , with R^2 values exceeding 0.9, while DML-Geo achieves the highest precision among them. For b_4 , however, GeoSHAPLEY exhibits substantial deviation from the true value, whereas both CausalForest and DML-Geo accurately reproduce it. These patterns indicate that the DML framework holds an advantage over non-causal models like GeoSHAPLEY in recovering linear effects. Note that the quadratic term X_4^2 must be explicitly designated as the treatment variable T to estimate b_4 in DML-Geo and CausalForest, whereas GeoSHAPLEY can incorporate higher-order terms directly in a single model fit via marginal attributions.

For Dataset 3, the inclusion of the interaction term X_1X_3 and the b_0X_0 term substantially increases its complexity. As shown in the Figure 4, STL consistently shows the largest estimation errors across all evaluations. Similarly, this complexity directly lowers the estimation accuracy of GeoSHAPLEY, particularly for b_0 and b_4 : the R^2 for b_0 estimation is only 0.4, while the estimation of b_4 exhibits a noticeable deviation near $X_4 = 0$. This

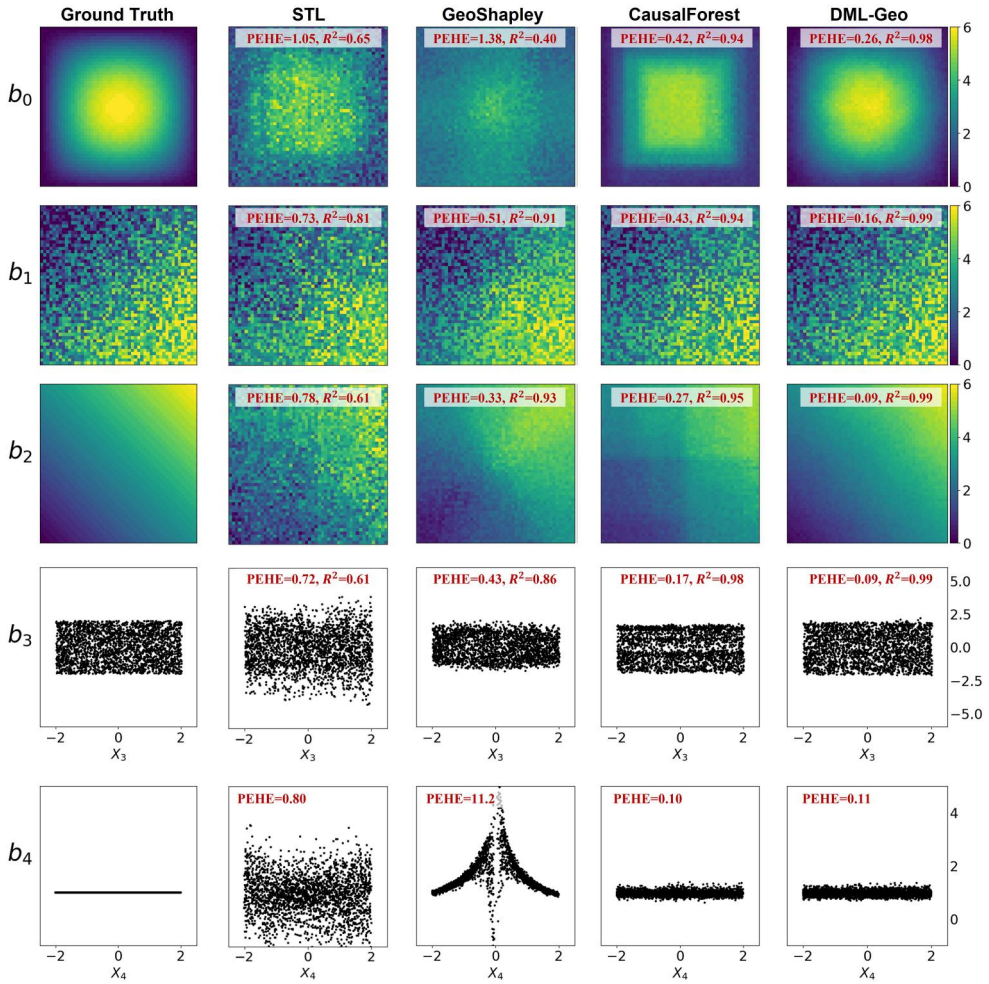


Figure 4. Comparison of estimated coefficients for Dataset 3 from different models. Notably, R^2 is undefined when the ground truth is constant.

likely stems from GeoSHAPLEY's reliance on marginal feature attributions, which fail to disentangle direct and indirect effects in the presence of correlated or interacting features (Kumar *et al.* 2021, Rozenfeld 2024). Although CausalForest outperforms GeoSHAPLEY, it displays blocky spatial patterns for b_0 and b_3 . This arises from its treatment of spatial coordinates as conventional covariates, without explicit incorporation of spatial dependencies. As a result, decision trees partition data exclusively along individual coordinate, making the estimated surfaces exhibit a pixelated, stepwise morphology rather than the smooth gradients typical of authentic spatial processes. In contrast, DML-Geo remains highly robust, achieving R^2 values all above 0.96 for the estimation of b_0 – b_4 .

As shown in Figure 5, further examination of the weights of component models across simulated datasets reveals a clear adaptive pattern: for causal effects exhibiting pronounced spatial patterns, such as b_1 , b_2 in Dataset 1 and b_0 in Dataset 3, DML-Geo assigns substantially higher weights to DML-GWR, while putting negligible weights on

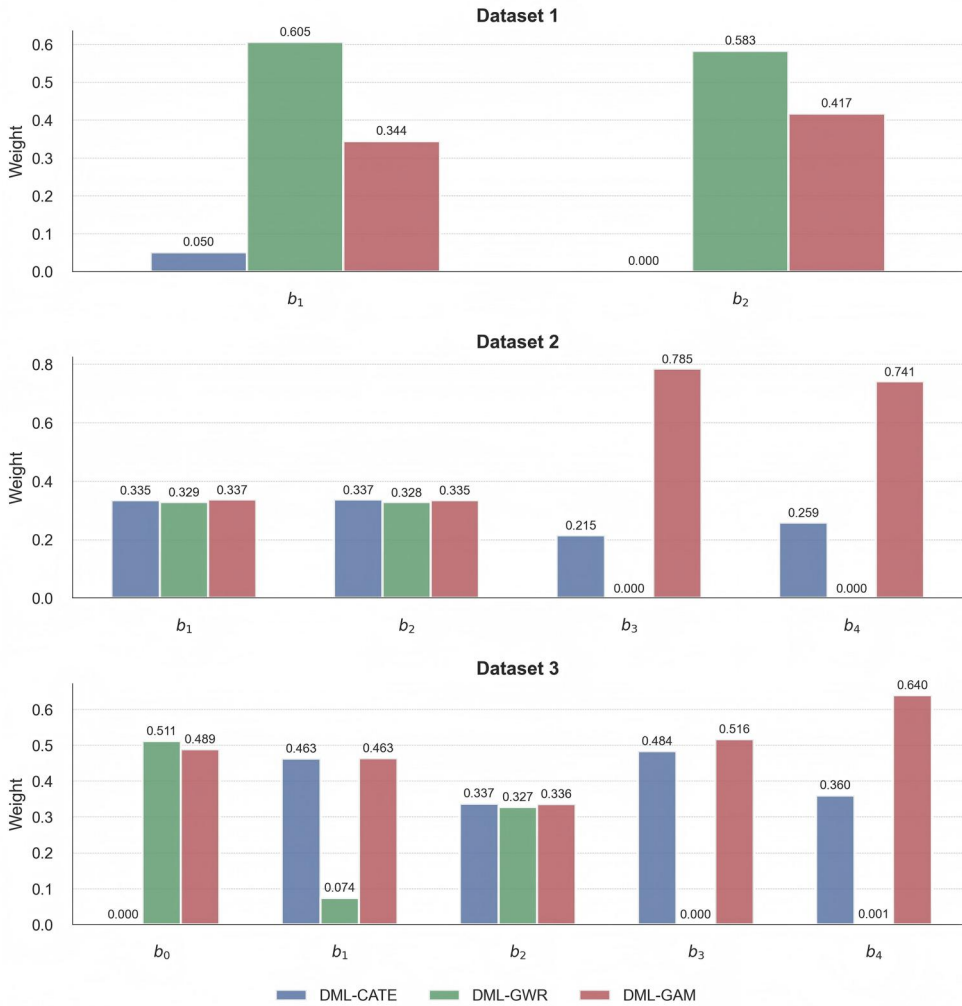


Figure 5. Weights of component models within DML-Geo across different simulated datasets.

DML-CATE. Conversely, for effects with negligible spatial signatures, such as b_3 , b_4 in Dataset 2 and b_1 , b_3 , b_4 in Dataset 3, DML-CATE receives elevated weights, with very small weights for DML-GWR. In contrast, DML-GAM exhibits non-negligible weights across scenarios. By accommodating both non-spatial heterogeneity and moderate spatial variations, DML-GAM acts as a balanced intermediary between DML-CATE and DML-GWR, playing a pivotal role as a harmonizing element within the ensemble. Overall, this demonstrates DML-Geo's capacity to adaptively prioritize the most reliable component models based on the underlying causal patterns

4.2. Extensive simulation across different scenarios

4.2.1. Construction of extensive simulated datasets

To assess the robustness of the DML-Geo framework and its variants under a broad spectrum of conditions, an extensive simulation study is conducted encompassing 118

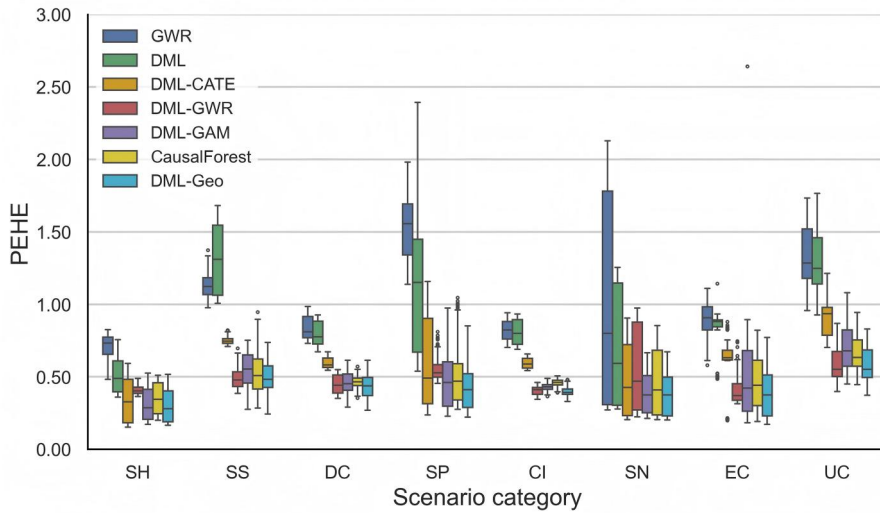


Figure 6. Comparison of model accuracy across different simulated scenarios.

synthetic datasets. These scenarios are organized into eight categories to probe specific sensitivities, including *spatial heterogeneity* (SH), *sample size sensitivity* (SS), *dimensionality challenge* (DC), *spatial pattern complexity* (SP), *confounding intensity* (CI), *strength of treatment effect and noise* (SN), *edge cases and stress tests* (EC), and *unobserved confounders* (UC). The detailed specifications for dataset generation and parameter configurations can be found in the [Supplemental Material](#) available via the Figshare repository.

Models evaluated include GWR, DML, DML-CATE, DML-GWR, DML-GAM, CausalForest and DML-Geo, with performance evaluated by PEHE in ATE/CATE recovery. The parameter α was initially set to 10 for DML-Geo. Notably, considering the randomness in synthetic data generation, we performed five independent replicates by varying random processes, yielding a total of $118 \times 5 = 590$ simulation datasets along with their corresponding causal inference results. This approach ensures more stable and reliable statistical results across diverse scenarios.

4.2.2. Comparison of different models

Figure 6 presents boxplots of PEHE in CATE recovery across the eight categories. As shown in Figure 6, the coefficients estimated by GWR deviate markedly from the true values, consistent with the principle that correlation does not imply causation (Pearl 2010). Among all causal inference models, the baseline DML performs the weakest, as it estimates only ATE. DML-CATE performs relatively well in cases with limited spatial heterogeneity (e.g., certain scenarios in SH, SP, and SN). However, in most settings where causal effects exhibit pronounced spatial variation, its performance deteriorates due to the absence of explicit spatial modeling. DML-GWR captures heterogeneity related to geographic coordinates. Consequently, although its accuracy is sometimes lower than that of DML-GAM, it shows greater stability (narrower interquartile ranges) in most scenarios except for SN. DML-GAM and CausalForest accommodate heterogeneity from both covariates and spatial location, generally yielding higher accuracy but

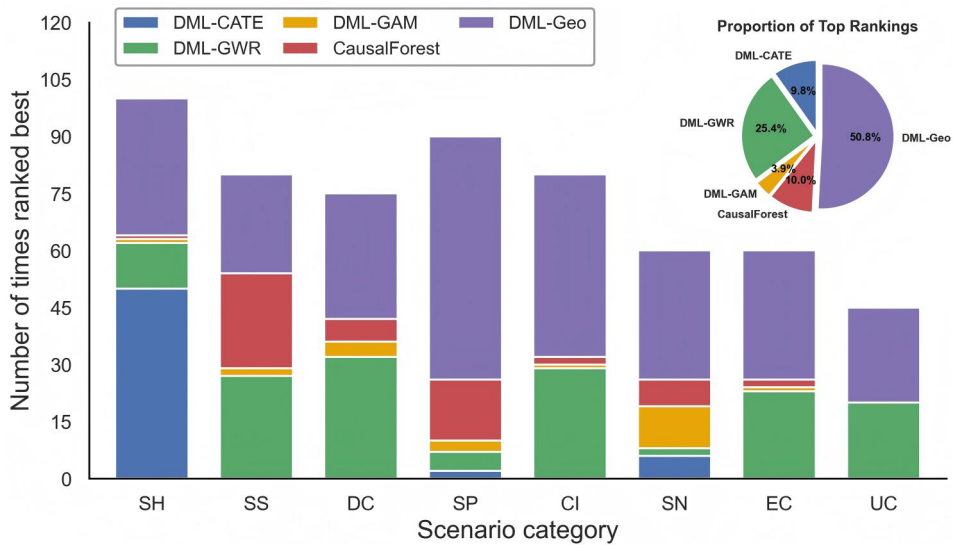


Figure 7. Frequency of each model achieving the lowest PEHE.

with less stability. DML-Geo adaptively combines results from DML-CATE, DML-GWR, and DML-GAM through weighted aggregation, achieving improved accuracy and stability across scenarios.

To further evaluate model performance, we counted, for each scenario category, the number of instances in which DML-CATE, DML-GWR, DML-GAM, CausalForest and DML-Geo achieved the lowest PEHE. As shown in the bar chart in Figure 7, DML-Geo attains a relatively high frequency of top rankings compared with all other models except SH and SS, indicating its strong overall accuracy. The pie chart in Figure 7 further shows that DML-Geo achieves the highest rank in 50.8% of the simulation scenarios, far exceeding the other methods. This highlights the higher robustness of DML-Geo than alternative models across diverse settings. We also note that although DML-GAM appears competitive in the boxplots of Figure 6, it rarely achieves the best performance. This is primarily because when DML-GAM performs well, DML-Geo can adaptively leverage it to improve its own results.

Based on extensive simulation experiments, we conducted pairwise comparisons of average percentage improvements in PEHE by designating each model as the baseline in turn. As depicted in Figure 8, DML-Geo consistently outperforms all the other approaches across scenarios. Relative to standard DML, it yields the largest average enhancement of 51.07%; against CausalForest, the average improvement reaches 11.98%; and even over DML-GAM, which is the closest contender, the gain remains notable at 8.60%.

4.2.3. Sensitivity analysis on the weighting parameter α

To explore a reasonable value for α in practice, we tested how model accuracy varied across different scenarios when α ranged from 1 to 20. As shown in Figure 9, in scenario like SH, DC and SN, the accuracy decreases as α increases, whereas in scenarios like SS, SP, CI, EC and UC, it improves initially and then declines. Considering all

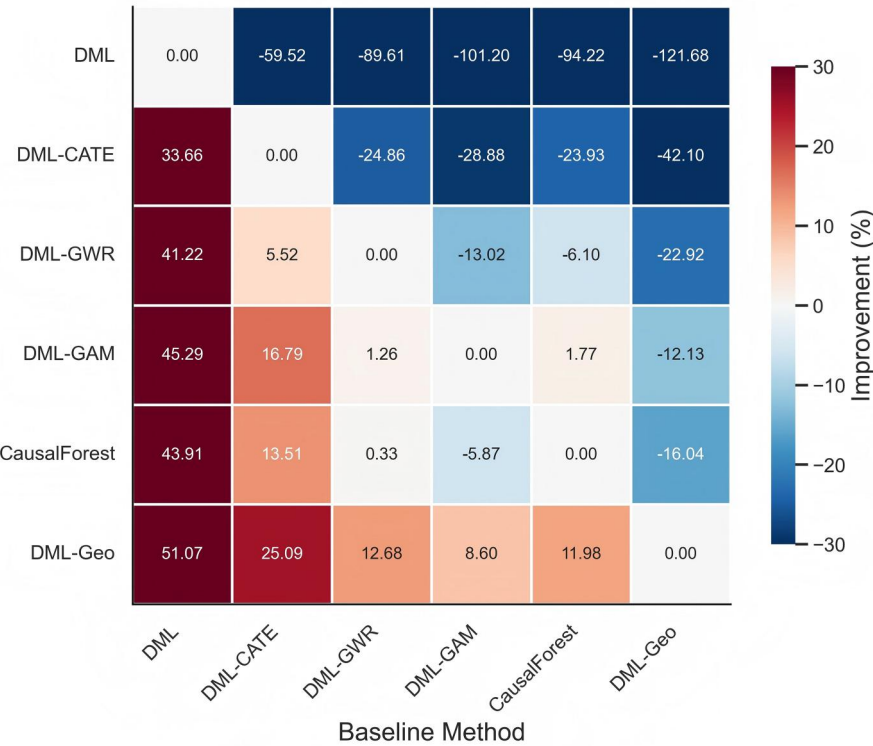


Figure 8. Pairwise average percentage improvements in PEHE across competing models from the simulation experiments. Heatmap values denote the relative enhancement of the row model over the column baseline, aggregated across all synthetic datasets.

simulation settings, $\alpha = 5$ yields relatively low PEHE across all scenario groups and is therefore adopted in the subsequent experiments on real-world datasets.

4.3. Experiments on real-world datasets

4.3.1. Data description

To further evaluate the proposed method, we employed two real-world datasets whose details are summarized in Table 1. The first dataset (House Price) covers house sales in King County, Washington, from May 2014 to May 2015, collected from GeoDa Lab (<https://geodacenter.github.io/data-and-lab/>). The original data contains over 20,000 records with 21 features related to property details, such as house sale prices, number of bedrooms, number of bathrooms, and living area in square feet. Following the preprocessing steps in (Li 2024a), we removed properties with more than ten bedrooms or lot sizes exceeding 20,000 square feet as outliers, and excluded records located in extremely remote areas based on their UTM coordinates after projection. In addition, we calculated the age and the adjusted age of each property based on their built and renovated years. We also excluded sqft_basement, as it sums exactly to sqft_living with sqft_above across all observations, which would induce perfect multi-collinearity in the first-stage nuisance regressions and preclude reliable second-stage

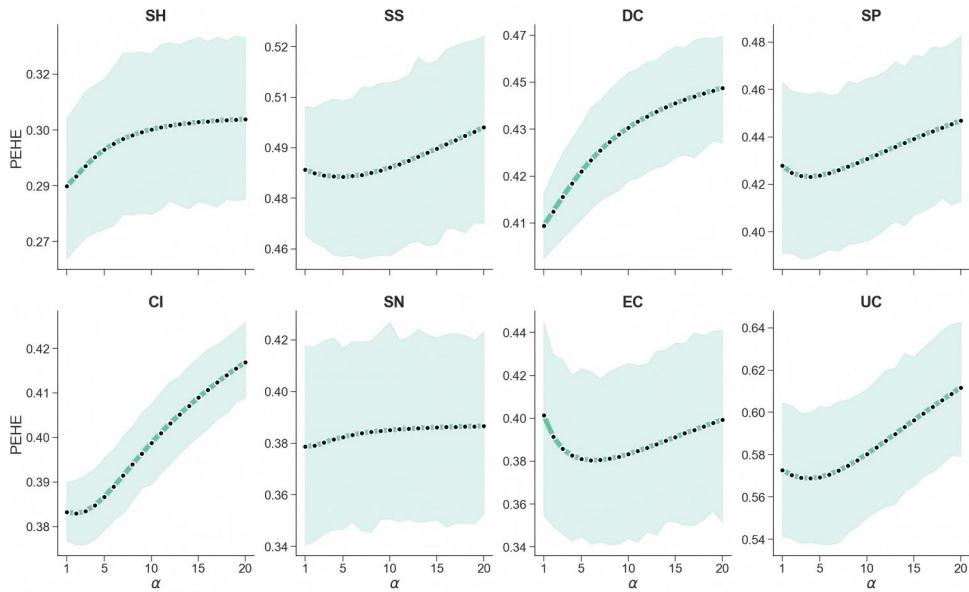


Figure 9. Sensitivity analysis of the parameter α on DML-Geo performance.

estimation. The final cleaned dataset contains 16,565 samples with 15 non-spatial features and two spatial coordinates.

The second dataset (Mental Health) was shared by Lessani and Li (2024) and accessible via the repository at (<https://github.com/Lessani252/SGWR>). It comprises 1,072 samples describing mental health prevalence across census tracts in South Carolina, USA, for the year 2019, with health data sourced from the Centers for Disease Control and Prevention (CDC) PLACES project. Each sample corresponds to a specific census tract and includes the mental health prevalence, 12 non-spatial predictor variables, and two spatial coordinates.

Particularly, we computed DML-Geo estimates independently for each feature to assess their CATE and further performed 100 bootstrap replications to evaluate significance. Notably, only results of DML-Geo are shown in the following content, while the results from the individual component models (e.g. DML-CATE, DML-GAM and DML-GWR), along with the proportions of significant effects and the corresponding weights, are detailed in Figure S1 of the Appendix.

4.3.1. Results on house price dataset

The spatial distribution of house price and the CATE estimates of various features are illustrated in Figure 10, where gray points indicate locations where the causal effects were deemed insignificant based on bootstrap testing. Several findings align closely with intuition. For instance, improvements in condition, grade, view, living area (*sqft_living*), lot area (*sqft_lot*) and the number of bathrooms generally lead to higher house prices. However, both the number of bedrooms and floors generally exert a negative influence on house prices in King County. This pattern arises because the county's housing stock is dominated by single-story detached homes, which command premiums over multi-story condominiums, thereby penalizing additional floors. In contrast,

Table 1. Summary of variables in the real-world datasets.

Dataset	Variable	Description	Transformation
House Price dataset	Dependent variable		
	price	House sale prices	Log
	Independent variables		
	grade	Construction quality classification based on materials and workmanship. Higher-grade buildings have higher construction costs and market values	None
	view	View quality index ranging from 0 to 4	None
	bathrooms	Number of bathrooms	None
	bedrooms	Number of bedrooms	None
	floors	Number of floors	None
	sqft_basement (discarded)	Basement area in square feet	Log
	sqft_living	Interior living space in square feet	Log
	sqft_lot	Total lot size in square feet	Log
	sqft_living15	Average living area of the 15 nearest neighboring houses (square feet)	Log
	sqft_lot15	Average lot size of the 15 nearest neighboring houses (square feet)	Log
	age	Property age as of 2015	None
	age_renovated	Years since last renovation (measured up to 2015)	None
	waterfront	Waterfront property indicator	None
Mental Health dataset	Dependent variable		
	Depression_Prevelence	Prevalence of diagnosed depression in the population.	None
	Independent variables		
	PopulationDensity	Density of the total population.	None
	PopulationFemale	Density of the female population.	None
	Population65	Density of the population aged 65 and over.	None
	NoHealthInsurance	Proportion of individuals without health insurance.	None
	Neighbor_Disadvantage	A composite index measuring the level of socioeconomic disadvantage in a neighborhood.	None
	Beer	Density of retail establishments related to beer (e.g., liquor stores).	None
	MentalHealthPati	Density of mental health service facilities.	None
	Generalhospitals	Density of general hospitals.	None
	NatureParks	Density of natural parks.	None
	Casinos	Density of casinos.	None
	DrinkingPlaces	Density of drinking places (e.g., bars).	None
	HouseRent	Proportion of households that are renter-occupied.	None

within Seattle's urban core, where multi-story condos predominate, higher floors may yield a modest positive premium due to enhanced views and natural light. Similarly, extra bedrooms often signal inefficient space utilization or elevated maintenance costs that fail to deliver commensurate value, particularly in dense urban environments.

Some features exhibit pronounced spatial heterogeneity. For example, in the western Lake Washington area, larger neighboring lots (*sqft_lot15*) appear to reduce prices. This is likely because this region is more densely developed, and larger lots correspond to lower neighborhood convenience and accessibility. By contrast, in residential communities like those in Bellevue, larger neighboring lots enhance privacy, outdoor space, and overall living comfort, leading to a positive effect on house prices. Unlike

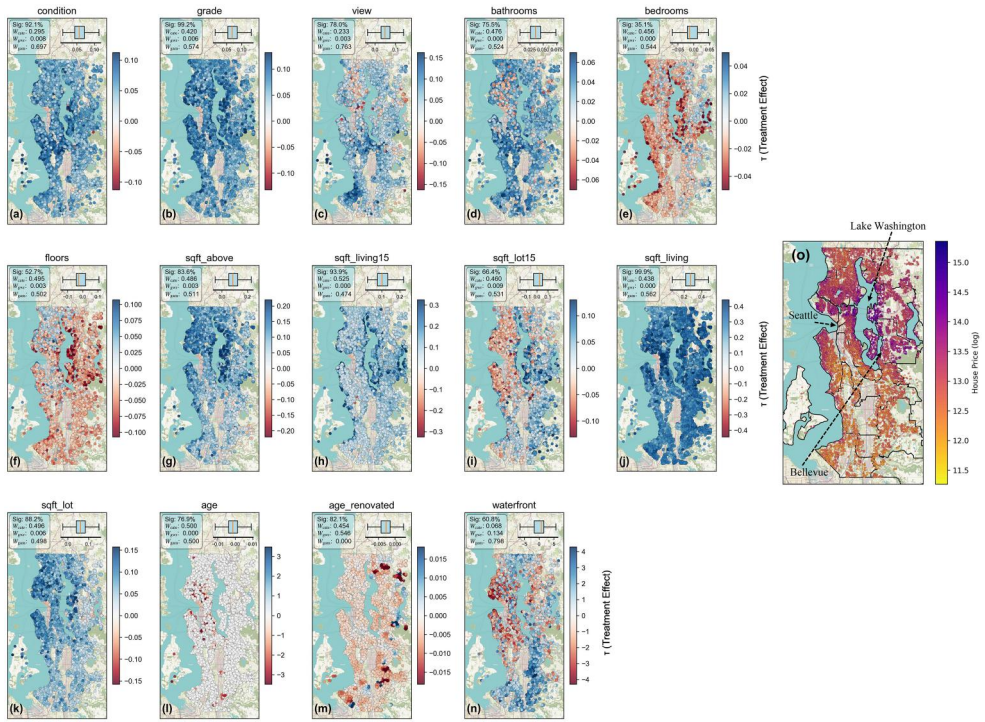


Figure 10. Map of House prices (o) and estimated causal effects for each variable (a–n). Boxplots of estimated effects are shown within each subplot, with data outside the whiskers being omitted.

sqft_lot15, neighboring living area (*sqft_living15*) generally shows a positive impact on house prices, reflecting the value of larger, higher-quality homes in the surrounding area. In addition, as shown in Figure 10(g), we found that *sqft_above* generally has a positive effect on house sale prices across most areas, yet negative effects emerge in the lower-right portion of the region. Further inspection shows that properties in this area typically lack basements. Given the additive relationship among *sqft_above*, *sqft_living*, and *sqft_basement*, this localized negative effect likely reflects that converting part of the above-ground area into a basement might enhance the home's functional utility and, in turn, improve its sale value.

Interestingly, in Figure 10(i), we found that property age exhibits a clear negative effect in a small number of regions, which is consistent with the expectation that older homes tend to sell for less. However, the corresponding boxplot reveals that, in many regions, property age shows a modest but significant positive effect on house sale price, which runs counter to expectations. Examination of the CATE for renovated age offers a compelling explanation for this counterintuitive result: more recent renovations appear to causally reduce house prices. This reflects the tendency for older properties to undergo renovations, thereby reducing their post-renovation age and boosting their market value. This finding is consistent with Li (2024). However, in conventional regression or SHAP analyses, renovated age is often excluded during preprocessing due to high variance inflation factors, which can lead to the counterintuitive conclusion that older homes command higher prices when age is assessed independently.

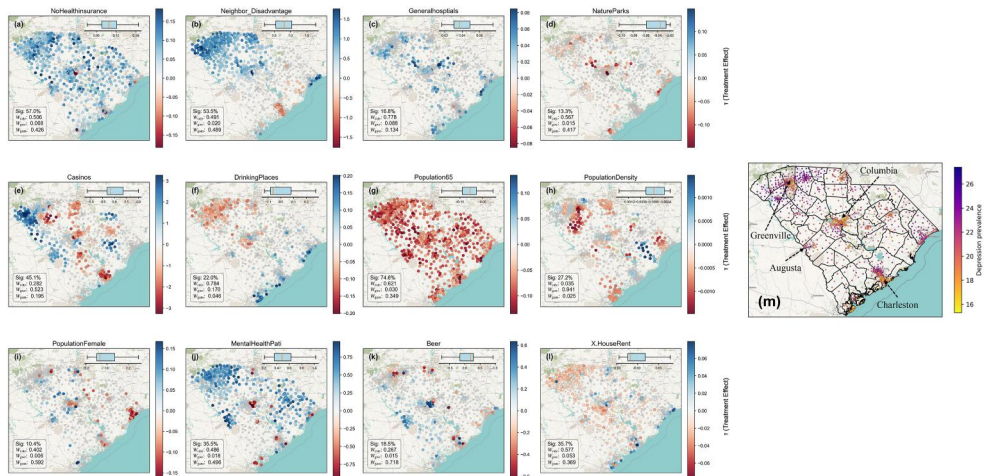


Figure 11. Map of Depression prevalence (m) and estimated causal effects for each variable (a–l). Boxplots of estimated effects are shown within each subplot, with data outside the whiskers being omitted.

Finally, the CATE results for *waterfront* (as shown in Figure 10(n)) indicate a mixed pattern: while waterfront locations generally increase property values in most inland areas, several zones around Lake Washington and Seattle’s coastal waterfront exhibit negative effects. Given the common expectation that water views enhance housing prices, these localized negative estimates appear counterintuitive. One possible explanation is that waterfront status may also capture environmental risks, such as increased exposure to flooding or higher maintenance requirements, which could offset the typical premium. However, we have reasons to suspect that this causal effect estimate may be unreliable: First, the DML-GWR estimates diverge substantially from those of the other two component models; Second, upon inspection, waterfront is a binary variable in this dataset, with only 88 positive instances (less than 0.5% of the total sample). Consequently, the extremely limited number of treated observations can lead to poor first-stage residual variation in DML and instability in machine learning estimates, undermining the reliability of the second-stage CATE estimation. This case illustrates a general recommendation: when counterintuitive results arise, practitioners should check for severe treatment imbalance and cross-validate component models, interpreting DML-Geo estimates with caution when substantial discrepancies exist.

4.3.2. Results on mental health dataset

Figure 11 illustrates the spatial distribution of mental health prevalence and the CATE values of various features. First, we observe that the proportion of significant causal effects for all the given factors remains below 60%, and the majority of non-significant points cluster in suburban areas, indicating that influential determinants in these regions may not have been fully captured by the observed variables. On a broader scale, several findings align with expectations: for instance, the proportion of residents without health insurance, neighborhood disadvantage, and the density of general hospitals show positive causal effects on mental health prevalence. In contrast, the presence of natural parks tends to exert a negative effect, potentially buffering against

psychological distress through enhanced access to restorative green spaces, although only about 17% of the CATE values are statistically significant.

Additionally, features linked to human behaviors, such as casino density and drinking-place density, show substantial spatial heterogeneity in their causal impacts. This variability highlights how the same exposure can yield divergent effects depending on local socioeconomic and cultural contexts. Specifically, the relationship between casino density and mental health prevalence varies in space. In mid-sized cities like Greenville and Augusta, where economic diversification and social support systems are comparatively limited, high casino concentrations can intensify financial strain and social instability, ultimately contributing to higher depression levels. In contrast, in larger and more diversified metros like Columbia and Charleston, casinos are typically embedded within broader entertainment economies, providing structured leisure options that may foster social engagement and mitigate psychological stress.

Similar patterns are observed for drinking-place density. In socially vibrant cities like Greenville and tourism-oriented cities like Charleston, a higher density of drinking establishments might serve as a hub for positive social interaction and community engagement, such as casual gatherings or cultural events, which can foster a sense of belonging and thereby mitigate psychological distress. In contrast, in Columbia, a hub of young adults and college students with high binge-drinking rates, drinking places amplify mental health risks through excessive alcohol consumption and related stress (Tembo *et al.* 2017). In surrounding suburbs, where social support is limited and at-risk populations are more scattered, these venues instead promote social isolation, substance abuse, and greater psychological distress.

Finally, an intriguing observation is that higher densities of individuals aged 65 and above are associated with lower depression prevalence. Cross-validation across the three component models shows strong consistency, supporting the reliability of these estimates (as shown in [Figure S2](#) of the Appendix). However, this global trend likely arises from underreporting of mental health issues among the elderly: older adults are less inclined to proactively seek mental health services, and CDC PLACES data, which are largely drawn from self-reported sources, may introduce bias and yield artificially low rates. In addition, depressive symptoms are also commonly dismissed as ‘normal aging’ or chronic illness effects. By contrast, total population density shows a significant but weak influence, whereas female population density shows a negative effect only in a few coastal regions and central Columbia. This may reflect the vibrant community activities in coastal and university-oriented locales, which cultivate stronger, more supportive social networks for women and thereby provide a protective buffer against psychological distress.

5. Discussion

The proposed DML-Geo framework represents a substantial advancement in causal inference for spatially heterogeneous effects within high-dimensional geospatial contexts. By integrating the flexibility of DML with different second-stage specifications and ensembling their individual outputs adaptively, DML-Geo effectively captures location-specific causal relationships that are often obscured in global

estimations. DML-Geo thus enables robust estimation of spatially varying causal effects while preserving interpretability. In the following, we discuss the model's key strengths, interpretative implications, strategies for addressing counterintuitive results, and remaining limitations.

A major strength of DML-Geo lies in its ability to accommodate numerous, potentially correlated spatial covariates, which is a persistent challenge in fields such as urban planning, environmental epidemiology, and real estate analytics. Traditional causal frameworks, constrained by parametric assumptions, often suffer from overfitting or omitted variable bias when faced with complex geospatial datasets. DML-Geo mitigates these issues through an orthogonalization process that employs flexible learners (e.g. random forests) to control for confounders, ensuring doubly robust estimation even under partial model misspecification. This property is particularly advantageous when working in geospatial studies, where spatial interaction and spatial autocorrelation are frequently present.

Equally important is the framework's streamlined approach to causal identification. Unlike instrumental variable or difference-in-differences designs, which rely on restrictive assumptions or hard-to-justify instruments, DML-Geo operates under the unconfoundedness condition, directly estimating causal effects conditional on observed covariates. This reduces dependence on subjective identification strategies and enables transparent, data-driven inference. In practice, this enables pointwise estimation of CATE surfaces without the procedural complexity of traditional econometric designs.

Another strength of DML-Geo lies in its ensemble-based stability. By aggregating component models (DML-CATE, DML-GAM, and DML-GWR) through correlation-based weighting, the framework reduces sensitivity to individual model specifications and mitigates biases that often affect single-model analyses. This produces more reliable CATE estimates under varying degrees of spatial heterogeneity. Bootstrap-based significance testing further ensures that reported spatial heterogeneity reflects genuine structural variation rather than estimation noise.

In terms of interpretability, DML-Geo provides detailed, map-based visualizations that show where and how treatment effects vary, enabling deeper insight into spatial mechanisms. For instance, the heterogeneous causal effects of above-ground living area on house sale prices in King County indicate variation in how residents value the functional features of a house. Conversely, counterintuitive findings, such as the negative effect of waterfront proximity or the seemingly protective effect of elderly population density on depression prevalence, underscore the need for contextual validation. We recommend a two-step interpretative protocol: first, verify consistency across component models; second, integrate domain knowledge to identify potential biases (e.g., underreporting in self-reported health surveys or sample imbalance in rare binary treatments). This protocol helps distinguish genuine causal heterogeneity from artifacts of model specification or data limitations, supporting more reliable inference.

Despite these strengths, DML-Geo faces some limitations. First, its nested learning and repeated bootstrapping incur high computational costs, especially for large spatial datasets. This limitation can be mitigated through parallelization, which distributes bootstrap replications or fold-wise model training across multiple CPU cores or GPUs,

and by adopting more efficient resampling schemes such as subsampling (m-out-of-n bootstrap). Second, the current ensemble is limited to three second-stage specifications (linear regression for baseline simplicity, GAM for smooth nonlinearities, and GWR for spatial localization), and does not explicitly account for spatial spillover effects from neighboring units. Future work could incorporate more advanced estimators such as spatial deep networks, or include neighboring treatments and attributes to better capture spillover dynamics across diverse geographic contexts. Lastly, the ensemble structure, while enhancing stability, may obscure the contribution of individual component models. We recommend component-wise visualization and sensitivity diagnostics. For example, mapping the spatial CATE patterns derived from DML-CATE, DML-GAM, and DML-GWR separately to identify consistent regions of effect and to trace potential sources of divergence.

6. Conclusion

This study presents DML-Geo, a framework specifically designed for spatial causal inference. By integrating three complementary DML variants targeting different forms of heterogeneity, DML-Geo can adaptively identify and quantify location-specific variations in causal effects. Its accuracy is first demonstrated on three representative simulated datasets and then comprehensively evaluated across 118 carefully designed simulation scenarios. The results indicate that DML-Geo achieves superior overall performance compared with competing methods, with an average 51% improvement in estimation accuracy relative to standard DML. From a practical perspective, DML-Geo offers an efficient and robust toolkit for spatial causal inference, capable of handling high-dimensional covariates and reducing reliance on detailed causal diagrams or strong prior structural assumptions. This enables both specialists and non-specialists to efficiently identify spatial causality for variables of interest. Future research should prioritize enhancing DML-Geo's computational efficiency, while also improving its accuracy through the integration of advanced DML variants (e.g. neural network-based methods). These extensions would enable the framework to tackle increasingly complex and diverse spatial causal inference challenges.

Author contributions

CRedit: **Pengfei Chen**: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Supervision, Visualization, Writing – original draft, Writing – review & editing; **Mengjie Gong**: Data curation, Formal analysis, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **Jingyu Wang**: Data curation, Formal analysis, Validation, Writing – review & editing; **Yixian Cai**: Data curation, Formal analysis, Investigation, Writing – review & editing; **Yiliang Wan**: Conceptualization, Funding acquisition, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Declaration of generative AI and AI-assisted technologies

In preparing this manuscript, we used generative AI tools in limited and transparent ways. ChatGPT (GPT-5.1) and Claude (Claude 3.7) were employed to assist in developing the initial design of the simulated data used in this study. After completing the full draft, ChatGPT (GPT-5.1) was additionally used for minor language refinement and grammar checking. All analytical decisions, methodological designs, and final interpretations were made solely by the authors.

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Data and code availability statement

The data, code, and supplemental material that support the findings of this study are available via the Figshare repository: <https://doi.org/10.6084/m9.figshare.30772610>

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